

Findings - Object Detection for Autonomous Vehicles

This document synthesizes the literature survey recorded in `sources.md`. All claims below are traceable to a specific source listed there; where I generalize across sources, the relevant section numbers from `sources.md` are cited in parentheses.

Scope and narrowing

“Object detection for autonomous vehicles” is too broad to survey exhaustively, so this survey was narrowed to six concrete use cases that, together, cover the perception stack of a real AV from sensor input to deployment constraint:

1. **2D camera-based real-time detection** - the baseline problem: find and classify cars, pedestrians, cyclists, signs, etc. in a single camera frame fast enough to drive on.
2. **3D / LiDAR point-cloud detection** - recovering depth and precise 3D location/size, which 2D image detection alone cannot provide.
3. **Multi-sensor fusion** (camera + LiDAR, and camera + radar as a noted adjacent area) - combining sensors to cover each one’s blind spots.
4. **Robustness under adverse weather** - rain, fog, night, snow - conditions where clean- weather-trained detectors degrade.
5. **Vulnerable road user (VRU) / pedestrian detection** - treated as a distinct sub-problem because of small object size, occlusion, and behavioral unpredictability.
6. **Edge/embedded deployment** - the constraint that all of the above must run in real time on limited, power-constrained, in-vehicle compute, not a data-center GPU.

Benchmark datasets (KITTI, nuScenes) are treated as a cross-cutting seventh area since almost every paper above is evaluated against them.

Development timeline observed across sources

Reconstructing publication order from the sources gathered:

- **2012** - KITTI benchmark suite (`sources.md` §4.1) establishes the first standardized, multi-task (2D/3D detection, stereo, flow, SLAM) driving benchmark, recorded from a single car with camera + LiDAR in one German city.
- **2016** - SSD (§1.1) demonstrates that single-stage detectors can match two-stage accuracy at real-time speed, a prerequisite for any onboard use.
- **2017–2019** - VoxelNet (§2.1) then PointPillars (§2.2) move 3D detection from hand-crafted point-cloud features to end-to-end learned representations, with PointPillars in particular hitting a speed/accuracy point usable in real time.

- **2019** - nuScenes (§3.2) generalizes the benchmark idea from KITTI to a full sensor suite (6 cameras, LiDAR, 5 radars) across 1,000 scenes, becoming the standard for fusion research; VRU-specific surveys (§6.1, §6.2) formalize pedestrian/cyclist detection as its own sub-field with intent estimation as the natural next step.
- **2020–2022** - Adverse-weather robustness becomes a dedicated research thread (rain review, §5.1); YOLOP (§1.2) and MobileYOLO (§1.3) show the shift toward multi-task and compressed-for-embedded detection, respectively; the camera+LiDAR fusion taxonomy survey (§3.1) organizes a fast-growing fusion literature.
- **2021–2024** - 3D detection survey (§2.3) takes stock of the field and explicitly adds runtime/robustness analysis as an evaluation axis alongside accuracy; adverse-weather work matures from “note the problem” to “engineer a fix” (YOLOv8 + merged-dataset fine-tuning, §5.3); edge-device benchmarking (§1.4) formalizes the accuracy/latency/ energy trade-off across concrete embedded hardware (Raspberry Pi, Jetson).

Read together, the field’s trajectory is: **establish a benchmark → get single-stage 2D detection fast enough → extend to 3D via LiDAR → fuse sensors → confront real-world robustness gaps (weather, VRUs) → compress/optimize for actual embedded hardware**. Each stage exists because the previous one is a precondition for it, not a random research trend - you cannot productively study weather robustness or embedded deployment for a detector that isn’t first real-time-capable and multi-sensor-aware.

Findings by use case

1. 2D real-time detection

One-stage detectors (SSD, YOLO family) dominate AV perception because AV detection is latency-critical: a two-stage detector’s separate region-proposal step adds sequential computation that one-stage detectors avoid by predicting boxes and classes directly from feature maps in a single pass (§1.1; see also the architecture diagram in §7.3). The trend within one-stage detection is now toward *multi-task* single networks - YOLOP (§1.2) performs detection + drivable-area segmentation + lane detection together, and still hits real-time speed on embedded Jetson hardware - because an AV needs all three outputs anyway, and sharing one encoder across tasks is cheaper than running three separate networks.

2. 3D / LiDAR detection

2D image detection cannot tell you how far away an object is or its true 3D extent, which is why LiDAR-based 3D detection is treated as a separate, necessary capability rather than a refinement of 2D detection (§2.3). The dominant architectural question in this area is how to encode a sparse, irregular point cloud into something a standard CNN can process efficiently: VoxelNet’s 3D voxel

grid (§2.1) was the first end-to-end learned answer, and PointPillars’ vertical-column (“pillar”) encoding (§2.2) proved that you can drop the expensive 3D convolutions and still get strong accuracy by projecting to a 2D pseudo-image - this pillar-based approach is why PointPillars remains a standard baseline in later benchmarks.

3. Multi-sensor fusion

No single sensor is sufficient: cameras give rich semantic detail but no reliable depth and degrade in poor light/weather; LiDAR gives precise 3D geometry but is sparse and expensive; radar is robust to weather but low-resolution. The key organizing question in fusion research is not “which sensors” but “at what stage do you combine them” - the camera+LiDAR fusion survey (§3.1) proposes a taxonomy based on fusion stage (raw data/early vs. feature-level vs. proposal/decision-level vs. hybrids) precisely because that design choice, more than sensor choice, determines a fusion architecture’s accuracy/latency/robustness trade-offs. The nuScenes dataset (§3.2) is the standard testbed for this area because it is the first large public dataset to include radar alongside camera and LiDAR.

4. Adverse weather robustness

Clean-weather-trained detectors reliably lose accuracy in rain, fog, snow, and low light (§5.1, §5.2) - this is treated in the literature as an expected, well-documented failure mode, not an edge case. Two mitigation strategies recur: (a) pre-process/enhance the image before detection (deraining, contrast restoration, domain adaptation/translation, per §5.1), or (b) fine-tune the detector directly on weather-degraded data, which the YOLOv8 + ACDC/DAWN dataset-merging study (§5.3) shows produces steady accuracy gains as more weather-diverse training data is merged in - i.e., dataset diversity is currently a more reliable lever than architectural changes for this specific problem.

5. VRU / pedestrian detection

Pedestrians and cyclists are harder to detect reliably than vehicles because they are smaller, more deformable, more often partially occluded, and behaviorally less predictable (§6.1). Because a detection alone doesn’t tell an AV whether a pedestrian is about to step into the road, the field has moved toward pairing detection with tracking, motion modeling, and pose-based intent estimation (§6.2) - detection is increasingly framed as the necessary first stage of a “detect → track → predict intent” pipeline rather than a standalone deliverable.

6. Edge/embedded deployment

Every architectural gain above (multi-task heads, pillar encodings, fusion) is constrained by what can actually run within an AV’s power and compute budget. The edge-device benchmarking study (§1.4) puts hard numbers on

this: SSD-MobileNetV1 is fastest and most energy-efficient but least accurate; YOLOv8-Medium is most accurate but most compute-hungry; hardware accelerators (Coral TPU) help lightweight models (SSD, EfficientDet-Lite) more than they help YOLOv8; and the Jetson Orin Nano currently offers the best overall accuracy/latency/energy balance among the tested embedded platforms. MobileYOLO (§1.3) is a model-level answer to the same pressure - a MobileNetv2 backbone plus depthwise-separable convolutions cut YOLOv4’s parameter count by over 50M while increasing detection speed by 70% on KITTI, illustrating the general recipe (lighter backbone + separable convolutions + lightweight attention) used across this literature to compress detectors for onboard hardware.

Cross-cutting observations

- **Benchmarks shape the field as much as architectures do.** KITTI (§4.1) and nuScenes (§3.2) are referenced by nearly every other paper surveyed; a new detection/fusion method is judged largely by how it compares on these two datasets. KITTI’s limitation - one city, daytime, clear weather - is part of why adverse-weather and VRU-specific research had to build (or reuse) additional datasets (ACDC, DAWN, per §5.3) rather than relying on KITTI alone.
- **Real-time constraint is the common thread across all six use cases.** Whether the topic is 2D detection, 3D detection, fusion, weather robustness, or VRU detection, nearly every paper surveyed reports a speed metric (FPS, ms latency, or explicit embedded-device benchmarking) alongside accuracy - accuracy without a latency number is treated as an incomplete result in this domain.
- **The “detect accurately” problem is gradually being absorbed into larger pipelines.** YOLOP folds detection into a shared multi-task network (§1.2); VRU research folds detection into a detect+track+predict-intent pipeline (§6.2); adverse-weather research folds detection into an enhance-then-detect or domain-adapt pipeline (§5.1, §5.3). Plain single-frame, single-task object detection is treated in current literature as a solved enough sub-problem that most active research composes it with something else, rather than trying to push single-task accuracy further in isolation.

Gaps and open challenges noted in the surveyed literature

- Occlusion and small/far-away object detection remain open problems for VRUs specifically (§6.1).
- Weather robustness solutions are mostly dataset- and fine-tuning-driven rather than architecturally solved, meaning performance is bounded by how representative the training data is of the deployment weather conditions (§5.1, §5.3).
- Embedded-hardware benchmarking (§1.4) shows accuracy still degrades measurably as scene complexity (object count) increases, even on the best-

performing hardware/model combinations - a concern for dense urban driving scenes specifically.

- Two MDPI-hosted reviews relevant to adverse weather (§5.2, §5.3 partially) and one MDPI/PMC-hosted paper on embedded YOLO variants (§1.3) could only be reviewed via their abstracts/introductions in this survey because their publishers block automated PDF downloads; a full read of these three papers would require manually downloading them through a browser (see `sources.md` §9 for the exact links and reasoning).

Folder contents summary

```
literature survey/  
  sources.md      <- full source log (this survey's paper trail)  
  findings.md     <- this document  
  papers/         <- 12 downloaded PDFs (see sources.md §1-6 for per-file details)  
  images/         <- 3 downloaded, openly licensed images (see sources.md §7)
```